EXPERIMENT-1

Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:
COLLEGE DATABASE:
STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

- **The ER diagram represents the College Database System. It is designed to store information about students, their classes, subjects, and internal assessment marks.**

1. **STUDENT Entity**
   The STUDENT entity stores the details of students studying in the college.

Attributes:

- USN (Primary Key)
- StudentName
- Username
- Address
- Phone No (shown as multi-valued attribute)
- Gender

2. **SEMSEC Entity**
   The SEMSEC entity represents Semester and Section details.

Attributes:

- SSID (Primary Key)
- SEM (Semester)
- SEC (Section)

3. **CLASS Relationship**
   The CLASS relationship connects STUDENT and SEMSEC.

- It represents that a student belongs to one semester and one section.
- The relationship is 1 : N

The relationship includes:

- USN
- SSID

4. **SUBJECT Entity**
   The SUBJECT entity stores details about subjects offered by the college.

Attributes:

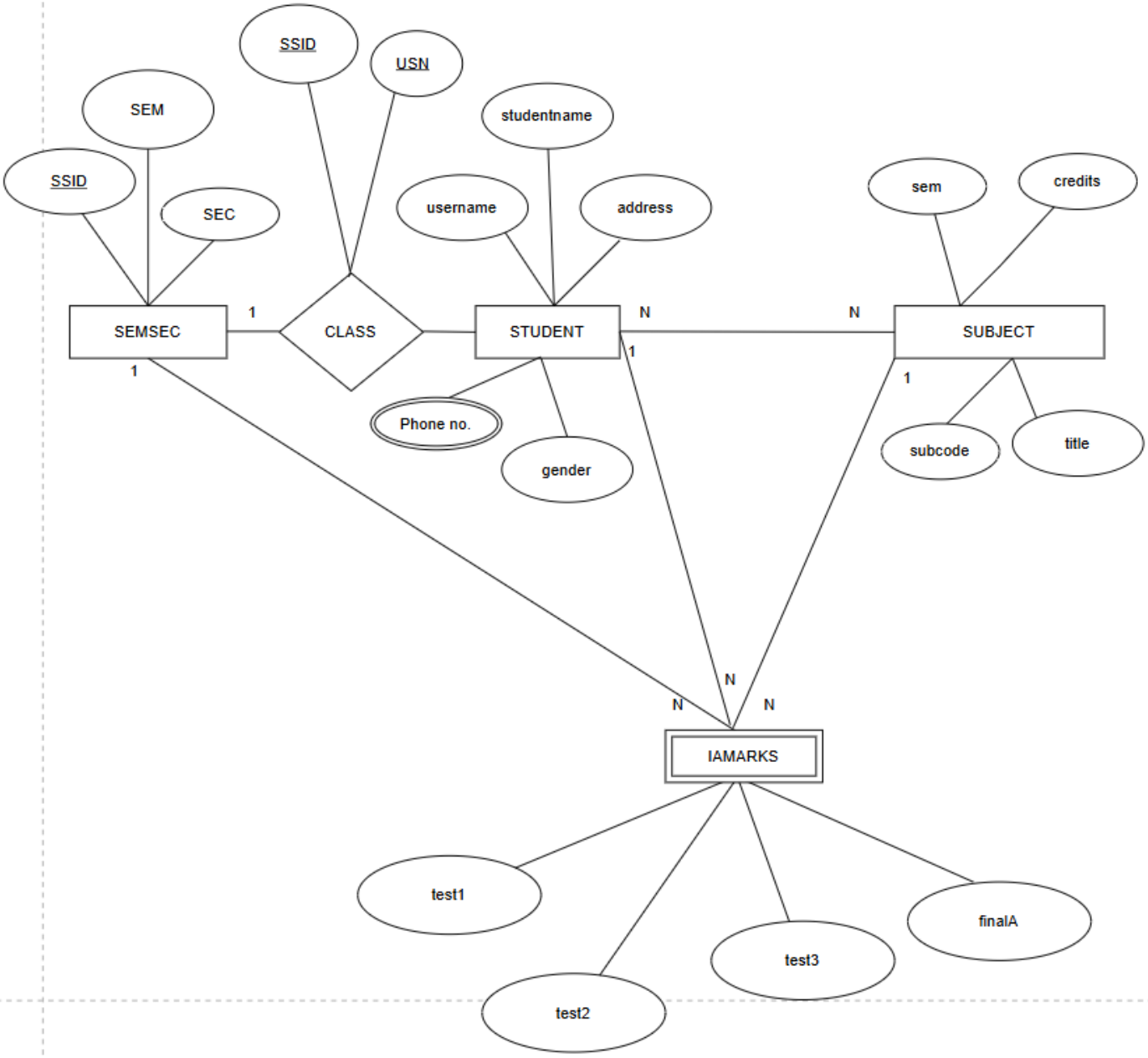
- Subcode (Primary Key)
- Title
- Semester
- Credits

5. **IAMARKS Entity**
   The IAMARKS entity stores Internal Assessment marks of students.

Attributes:

- USN
- Subcode
- SSID
- Test1
- Test2
- Test3
- FinalIA

ER DIAGRAM


SSID
USN
SEM
studentname
SSID
SEC
username
address
sem
credits
SEMSEC
1
CLASS
STUDENT
N
N
SUBJECT
1
1
1
Phone no.
gender
subcode
title
N
N
N
IAMARKS
test1
test2
test3
finalA

EXPERIMENT 2

Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:
COMPANY DATABASE:
EMPLOYEE (SSN, Name, Address, Sex, Salary, SuperSSN, DNo)
DEPARTMENT (DNo, DName, MgrSSN, MgrStartDate)
DLOCATION (DNo,DLoc)
PROJECT (PNo, PName, PLocation, DNo)
WORKS_ON (SSN, PNo, Hours)

COMPANY DATABASE: The given ER diagram represents a Company Database System. It describes the structure of the organization by showing entities, their attributes, and the relationships among them.

1. DEPARTMENT Entity
The **DEPARTMENT** entity stores information about departments in the company.
Attributes:
- **DNO** (Primary Key)
- DNAME
- MgrSSN (Manager SSN)
- MgrStartDate

2. EMPLOYEE Entity
The **EMPLOYEE** entity stores employee details.
Attributes:
- **SSN** (Primary Key)
- Name
- Sex
- Address
- Salary
- SuperSSN (Supervisor SSN)
- DNO (Department Number – Foreign Key)

3. PROJECT Entity
The **PROJECT** entity contains details about company projects.
Attributes:
- **PNO** (Primary Key)
- PNAME
- PLOCATION
- DNO (Department Number – Foreign Key)

4. DLOCATION Entity
The **DLOCATION** entity represents department locations.
Attributes:
- DNO (Foreign Key)
- DLOC

5. MANAGES Relationship
The **MANAGES** relationship connects:
- EMPLOYEE
- DEPARTMENT

Cardinality:
- One department is managed by one employee.
- An employee may manage one department.

(1 : 1 relationship)

6. CONTROLS Relationship
The **CONTROLS** relationship connects:
- DEPARTMENT
- PROJECT

Cardinality:
- One department controls many projects.
- Each project is controlled by one department.

(1 : M relationship)

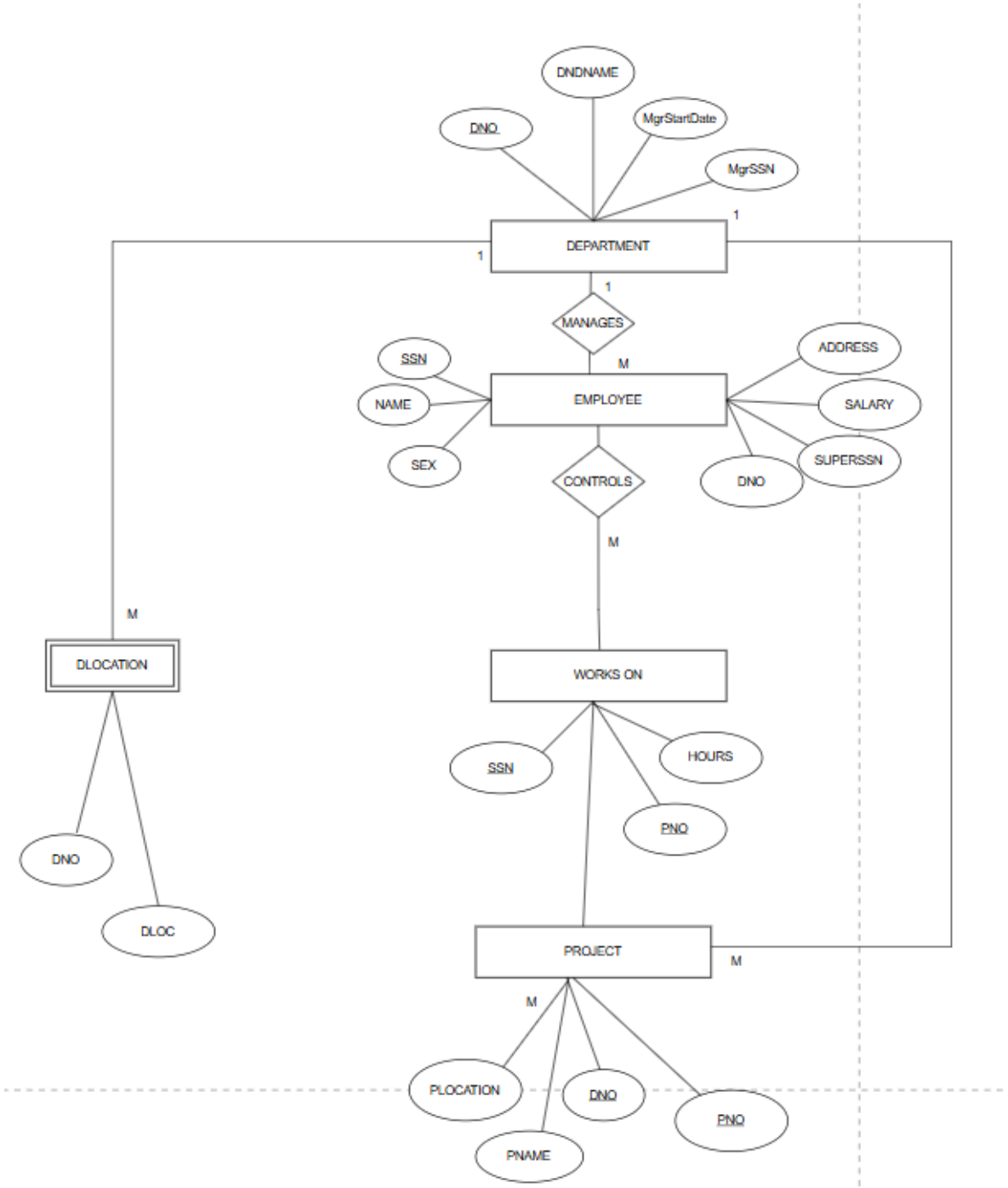
7. WORKS_ON Relationship
The **WORKS_ON** relationship connects:
- EMPLOYEE
- PROJECT

Attributes:
- SSN
- PNO
- Hours

Cardinality:
- One employee can work on many projects.
- One project can have many employees.

(M : N relationship)

ER DIAGRAM


DNDNAME
DNO
MgrStartDate
MgrSSN
DEPARTMENT
1
1
1
MANAGES
M
SSN
NAME
SEX
EMPLOYEE
ADDRESS
SALARY
SUPERSSN
DNO
CONTROLS
M
M
DLOCATION
WORKS ON
SSN
HOURS
PNO
DNO
DLOC
PROJECT
M
M
PLOCATION
DNO
PNO
PNAME